Seatwork No: 6.2	
Activity Title: Built-in Functions	
Course Code: CPE 007	**Program:** Computer Engineering
Course Title: Programming Logic and Design	**Date Performed:** 10/22/26
Section: CPE11S2	**Date Submitted:** 10/22/26
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Objectives

This activity aims to develop programming skills using functions.

Instructions

1. Create a program that defines a function to compute for the area of a cube. The formula of the area of the cube is given as A = s * s * s.
2. Define a function **hypotenuse** that calculates the length of the hypotenuse of a right triangle when the other two sides are given. Use this function in a program to determine the length of the hypotenuse for each of the following triangles. The function takes two arguments of type **double** and return the hypotenuse as a **double.**
3. Implement the following integer functions:
    1. Function **celsius** returns the Celsius equivalent of a Fahrenheit temperature.
    2. Function **fahrenheit** returns the Fahrenheit equivalent of a Celsius temperature.
    3. Use these functions to write a program that prints charts showing the Fahrenheit equivalents of a Celsius temperatures from 0 to 100 degrees, and the Celsius equivalents of all Fahrenheit temperatures from 32 to 212 degrees. Print the outputs in a neat tabular format that minimizes the number of lines of output while remaining readable.

Output/Answer

ACTIVITY 1:

Code:

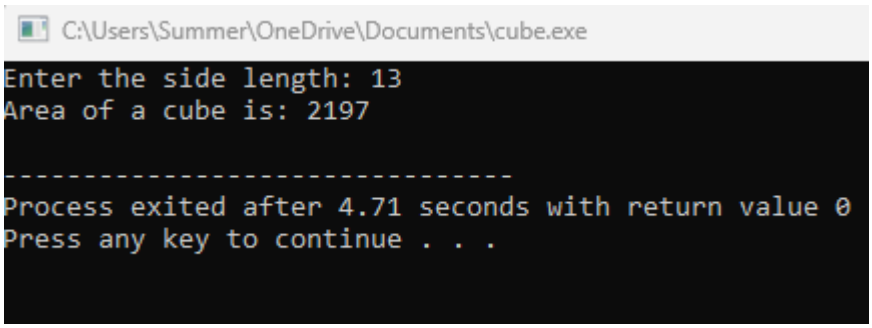
```
#include <iostream>
#include <cmath>
using namespace std;

// in the case of all sides being the same and we're calculating the volume, we used pow(x,y) to calculate for the all the three sides multiplied.
double area(double  x) {
        // since pow is x^y, we put y as 3 because it's s * s * s, x being the s being multiplied from itself.
        return pow(x, 3);
}

int main () {
        double a;
        // since all sides are the same in a cube, we need to input a given side
        cout << "Enter the side length: ";
        cin >> a;

        // we recalled the function pow
        int cubeA = area(a);
        cout << "Area of a cube is: " << cubeA << endl;
```

```
        return 0;
}
```



ACTIVITY 2:

Code:

```
#include <iostream>
#include <cmath>
using namespace std;

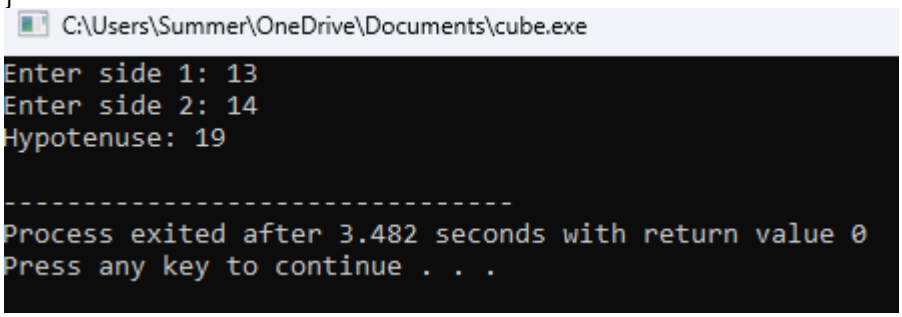
// same concept as the pow() once except we use the sqrt() library function wherein we find the square root of x. wherein the formula of a hypotenuse is sqrt of a^2 + b^2.
int hypotenuse (int side1, int side2) {
        return sqrt (side1 * side1 + side2 * side2);
}

int main() {
        int x, y, hypo;
        cout << "Enter side 1: ";
        cin >> x;

        cout << "Enter side 2: ";
        cin >> y;

        hypo = hypotenuse(x, y);
        cout << "Hypotenuse: " << hypo << endl;

        return 0;
}
```



ACTIVITY NO. 3

CODE:

```
#include <iostream>
#include <cmath>
using namespace std;

double celsius(double fahrenheit1) {
double five = pow (5, 1);
double nine = pow (9,1 );
return (five/nine) * fabs(fahrenheit1  - 32);
}

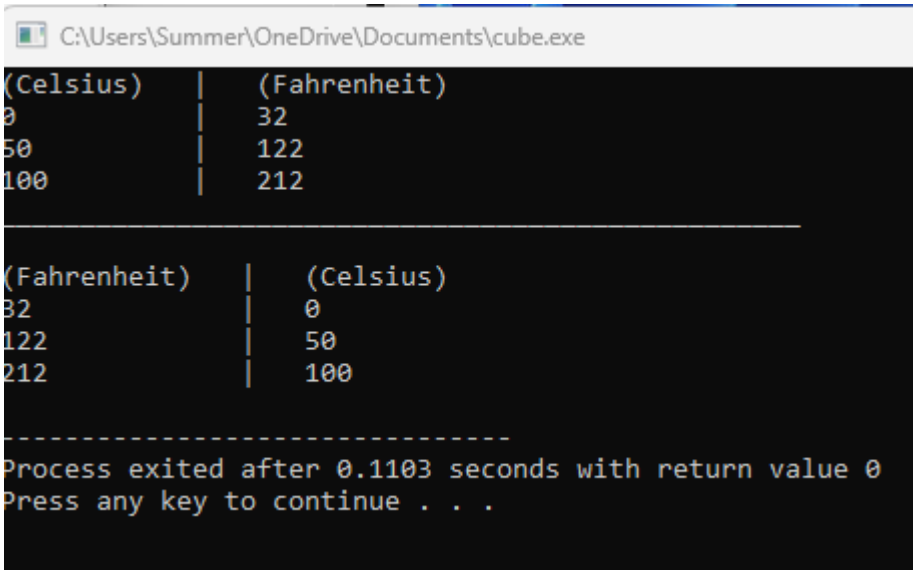
double fahrenheit (double celsius1) {
        double nine = pow(9, 1);
        double five = pow(5, 1);
        return (nine/five) * celsius1 + 32;
}
int main() {

        double a = 0;
        double b = 50;
        double c = 100;


        cout << "(Celsius)   " << "|" << "   (Fahrenheit)" << endl;
        cout << a << "           " << "|" << "   " << fahrenheit(a) << endl;
        cout << b << "          " << "|" << "   " << fahrenheit(b) << endl;
        cout << c << "         " << "|" << "   " << fahrenheit(c) << endl;

        cout << "_____"  << endl;
        cout << "                                          " << endl;

        cout << "(Fahrenheit)   " << "|" << "   (Celsius)" << endl;
        cout << fahrenheit(a) << "             " << "|" << "   " << celsius(fahrenheit(a)) << endl;
        cout << fahrenheit(b) << "            " << "|" << "   " << celsius(fahrenheit(b)) << endl;
        cout << fahrenheit(c) << "            " << "|" << "   " << celsius(fahrenheit(c)) << endl;
        return 0;
}
```



Supplementary Activity (if any)

Conclusion

Difficult but manageable.

Assessment Rubric: SO 7

Rubric for SO 7 (12)

Criteria	Ratings						
SO 7 PI 1 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6
SO 7 PI 2 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently	6
SO 7 PI 3 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6
SO 7 PI 4 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted	6

Total Points: